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Problem set #3

Submission Date: 02/24/2026

Due Date: 02/24/2026

Acknowledgement

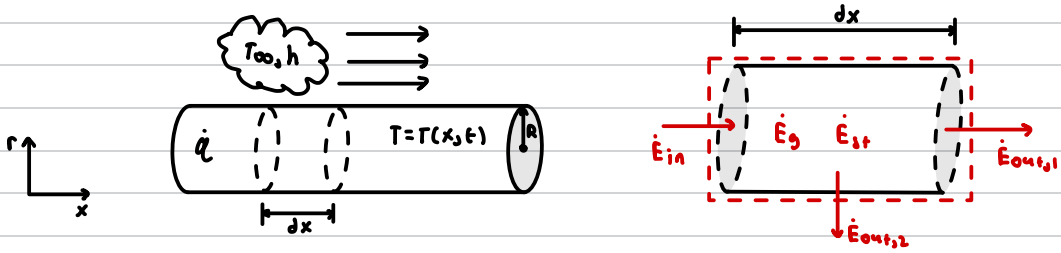
- reffered to slide of lecture 6, 7, and 8 slides

1) **Known:** R, $\dot{q}$, T_∞, h, Boundary Initial conditions

Find: a) Energy balance for CV of differential cylinder (dx). Express each energy term as function of T.

b) Governing DE describing temp profile

Schematic



Assumptions:

- Thermal radiation can be neglected
- $R \ll L$
- $\frac{\partial T}{\partial r} = 0$, $\nabla k = \nabla \rho = \nabla c_p = \vec{0}$

Analysis

a) Volume Enegy Balance: $\dot{E}_{in} + \dot{E}_g - \dot{E}_{out,1} - \dot{E}_{out,2} = \dot{E}_{st}$ Area: $A = \pi R^2$ Perimeter: $P = 2\pi R$

$\dot{E}_{in} = q_x = -k_x \frac{\partial T}{\partial x} A$

$\dot{E}_g = \dot{q} A dx$

$\dot{E}_{out,1} = q_{x+dx} = q_x + \frac{\partial q_x}{\partial x} dx = -k_x \frac{\partial T}{\partial x} A + \frac{\partial}{\partial x}\left(-k_x \frac{\partial T}{\partial x} A\right)$

$\dot{E}_{out,2} = q_{conv} = hP(T - T_\infty) dx$

$\dot{E}_{st} = \rho c_p \frac{\partial T}{\partial t} A dx$

b) $q_x + \dot{q} A dx - q_{x+dx} - hP(T - T_\infty) dx = \rho c_p \frac{\partial T}{\partial t} A dx$

$q_x - q_x - \frac{\partial q_x}{\partial x} dx + \dot{q} A dx - hP(T - T_\infty) dx = \rho c_p \frac{\partial T}{\partial t} A dx$

$\frac{\partial}{\partial x}\left(k_x \frac{\partial T}{\partial x}\right) A dx + \dot{q} A dx - hP(T - T_\infty) dx = \rho c_p \frac{\partial T}{\partial t} A dx$

$\frac{\partial}{\partial x}\left(k_x \frac{\partial T}{\partial x}\right) \pi R^2 + \dot{q} \pi R^2 - h 2\pi R (T - T_\infty) = \rho c_p \pi R^2 \frac{\partial T}{\partial t}$ $\frac{\partial}{\partial x}\left(k_x \frac{\partial T}{\partial x}\right) + \dot{q} - \frac{2h}{R}(T - T_\infty) = \rho c_p \frac{\partial T}{\partial t}$

c) Boundry at $x = 0$: $T(0,t) = T_1$

Boundry at $x = L$: $-k_x \frac{\partial T}{\partial x}\Big|_{x=L} = h(T(L,t) - T_\infty)$

Initial conditions: $T(x,0) = T_i$

a) $\dot{E}_{in} + \dot{E}_g - \dot{E}_{out,1} - \dot{E}_{out,2} = \dot{E}_{st}$

$\dot{E}_{in} = -k\pi R^2 \frac{\partial T}{\partial x}$

$\dot{E}_g = \dot{q} \pi R^2 dx$

$\dot{E}_{out,1} = -k\pi R^2 \left(\frac{\partial^2 T}{\partial x^2} + \frac{\partial T}{\partial x}\right)$

$E_{out,2} = h 2\pi R (T - T_\infty) dx$

$\dot{E}_{st} = \rho c_p \pi R^2 \frac{\partial T}{\partial t} dx$

b) $\frac{\partial}{\partial x}\left(k_x \frac{\partial T}{\partial x}\right) + \dot{q} - \frac{2h}{R}(T - T_\infty) = \rho c_p \frac{\partial T}{\partial t}$

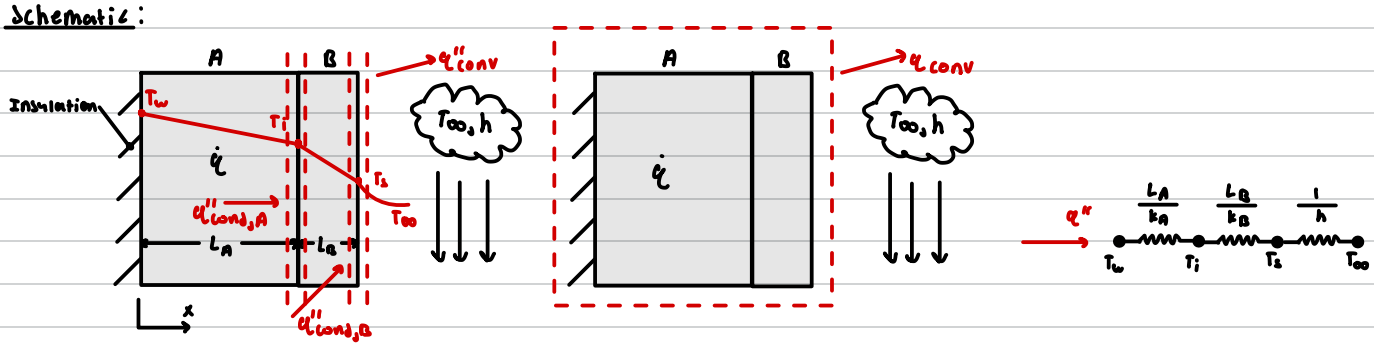
c) $T(0,t) = T_1$ $-k \frac{\partial T}{\partial x}\Big|_{x=L} = h[T(L,0) - T_\infty]$

$T(x,0) = T_i$

2) **known**: $L_A = 30cm$, $\dot{q}$, k_A, $\frac{\partial T}{\partial x}\Big|_{x=0} = 0$, $L_B = 10cm$, $k_B = 30 \frac{W}{mK}$, $T_\infty = 20°C$, $h = 500 \frac{W}{m^2K}$, $T_w = 115°C$, $T_i = 100°C$

Find: a) calculate T_s of wall B
b) calculate $\dot{q}$ at steady state
c) calculate k_A
d) Plot temperature profile across wall A and wall B

Schematic:



Assumption:
- Steady state
- 1D heat conduction
- No contact thermal resistor at wall A and wall B boundary
- Negligible thermal radiation

Analysis

a) $q'' = \frac{T_1 - T_2}{R''_{tot}}$ $\quad q''_{cond,B} = T_i - T_\infty \Big/ \frac{L_B}{k_B} + \frac{1}{h} = (100°C - 20°C) \Big/ \frac{0.1m}{30 W/mK} + \frac{1}{500 W/m^2K} = 15 \frac{kW}{m^2}$

Surface Energy Balance: $\dot{E}_{in} = \dot{E}_{out} \longrightarrow q''_{conv,B} = q''_{cond} = h(T_s - T_\infty)$

$$T_s = T_\infty + \frac{q''_{conv}}{h} = 20°C + \frac{15000 W/m^2}{500 W/m^2K} = 50°C$$

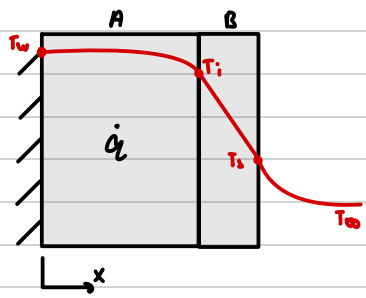
b) Volume Energy Balance: $\dot{E}_{in} + \dot{E}_g = \dot{E}_{out} \longrightarrow \dot{q}V = q''_{conv}A = q''_{cond,B}A$

$$\dot{q}A(L_A + L_B) = q''_{conv}A \quad \dot{q} = \frac{q''_{cond,B}}{L_A} = \frac{15000 W/m^2}{0.30m} = 50 \frac{kW}{m^2}$$

c) $\frac{\partial}{\partial x}\left(k_x \frac{\partial T}{\partial x}\right) + \dot{q} = \rho c_p \frac{\partial T}{\partial t}$ $\quad k\frac{\partial^2 T}{\partial x^2} + \dot{q} = 0$ $\quad k\frac{\partial T}{\partial x} + \dot{q}x + a = 0$ $\quad \frac{\partial T}{\partial x} = -\frac{\dot{q}}{k}x + C$ $\quad T(x) = -\frac{\dot{q}}{2k}x^2 + ax + b$

$\frac{\partial T}{\partial x}\Big|_{x=0} = 0 \rightarrow a = 0$ $\quad T(0) = T_w \rightarrow b = T_w$ $\quad T(x) = T_w - \frac{\dot{q}}{2k}x^2$ $\quad k = \frac{\dot{q}x^2}{2(T_w - T(0))} = \frac{\dot{q}L_A^2}{2(T_w - T_i)} = \frac{(50 kW/m^3)(0.30m)^2}{2(115°C - 100°C)} = 150 \frac{W}{mK}$

d)



$$\boxed{a)\ T_s = 50°C \quad b)\ \dot{q} = 50 \frac{kW}{m^3} \quad c)\ k_A = 150 \frac{W}{mK}}$$

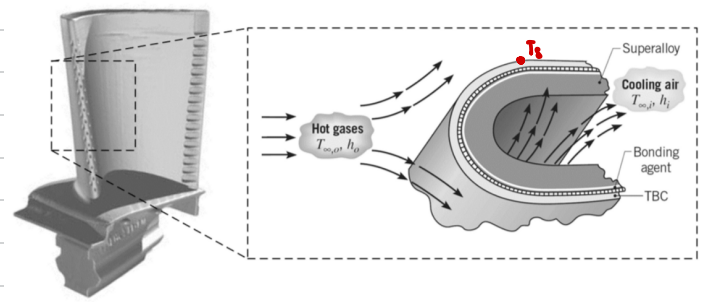
Temp profile in A is parabolic with a high k value (small temp difference).
Temp profile in B is linear with a small k value (big temp difference).

3. **Given:** $k_s = 25\ \frac{W}{mK}$, $k_{TBC} = 1\ \frac{W}{mK}$, $R'' = 10^{-4}\ \frac{m^2K}{W}$, $T_{\infty,o} = 1616K$, $T_{\infty,i} = 400K$, $h_o = 1000\ \frac{W}{m^2K}$, $h_i = 500\ \frac{W}{m^2K}$, $L_b = 5mm$, $T_{max} = 1200K$

Find: a) no TBC applied. Draw thermal resistance network. calculate T_s. determine whether $T_{s,o} < T_{max}$

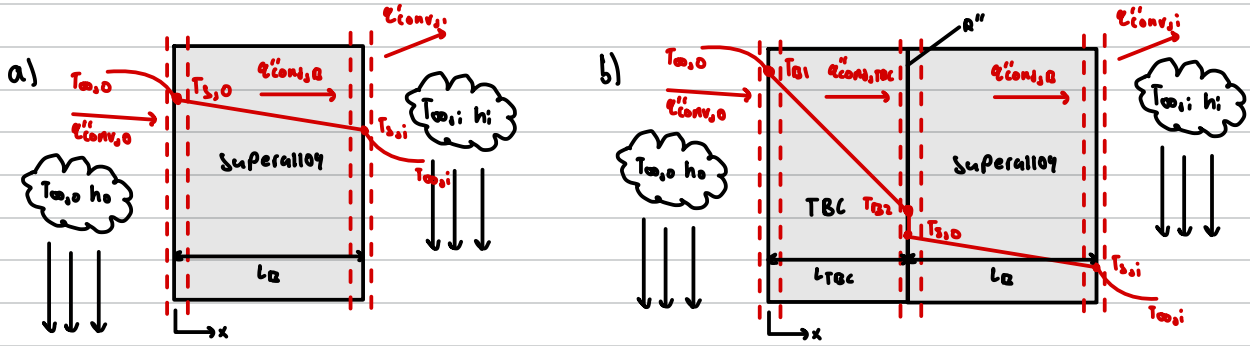
b) $L_{TBC} = 0.5mm$. Draw thermal resistance network. calculate T_s. determine whether $T_{s,o} < T_{max}$

Schematic



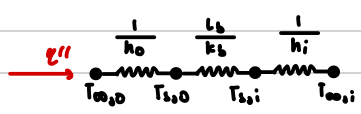
Assumptions

- steady state
- negligable thermal radiation
- turbin blade thin wall aproximation
  ↳ 1D heat conduction through TBC and superalloy



Analysis

a)



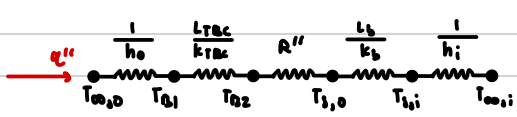
$$q'' = \frac{T_1 - T_2}{R''_{tot}}$$

$$R''_{tot} = \frac{1}{h_o} + \frac{L_b}{k_b} + \frac{1}{h_i} = \frac{1}{1000\ W/m^2K} + \frac{0.005m}{25\ W/mK} + \frac{1}{500\ W/m^2K} = 32\times10^{-4}\ \frac{m^2K}{W}$$

$$q'' = (T_{\infty,o} - T_{\infty,i})/R''_{tot} = \frac{(1616K - 400K)}{32\times10^{-4}\ m^2K/W} = 380\ \frac{kW}{m^2}$$

$$q'' = \frac{T_{\infty,o} - T_{s,o}}{R''_{h_o}} = h(T_{\infty,o} - T_{s,o}) \qquad T_s = T_\infty - \frac{q''_{cond,b}}{h} = 1616K - \frac{380000\ W/m^2}{1000\ W/m^2K} = 1236K$$

b)



$$q'' = \frac{T_1 - T_2}{R''_{tot}}$$

$$R''_{tot} = \frac{1}{h_o} + \frac{L_{TBC}}{k_{TBC}} + R'' + \frac{L_b}{k_b} + \frac{1}{h_i} = \frac{1}{1000\ W/m^2K} + \frac{0.0005m}{1\ W/mK} + 10^{-4}\ \frac{m^2K}{W} + \frac{0.005m}{25\ W/mK} + \frac{1}{500\ W/m^2K} = 38\times10^{-4}\ \frac{m^2K}{W}$$

$$q'' = (T_{\infty,o} - T_{\infty,i})/R''_{tot} = \frac{(1616K - 400K)}{38\times10^{-4}\ m^2K/W} = 320\ \frac{kW}{m^2}$$

$$q'' = \frac{T_{\infty,o} - T_{s,o}}{R''_{h_o} + R''_{k_{TBC}} + R''} \qquad T_{s,o} = T_{\infty,o} - q''\left(\frac{1}{h_o} + \frac{L_{TBC}}{k_{TBC}} + R''\right) = 1616K - 320000\ \frac{W}{m^2K}\left(\frac{1}{1000\ W/m^2K} + \frac{0.0005m}{1\ W/m^2K} + 10^{-4}\ \frac{W}{m^2K}\right) = 1104K$$

$$\boxed{\begin{array}{l} a)\ T_{s,o} = 1236K > T_{max} \\ b)\ T_{s,o} = 1104K < T_{max} \end{array}}$$